

Liability Driven Immunization & Credit Risk Portfolio Analysis

FRL 6460

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Introduction

This project evaluates how effective liability-driven fixed income immunization strategies are when implemented using both investment grade and high yield credit instruments. This project covers the period from December 2018 through December 2025 and focuses on constructing portfolios capable of meeting a predetermined liability of \$100 million due at the end of 2025.

This project analyzed the real world performance of fixed income immunization strategies under evolving market conditions using historical Bloomberg data. The study focused on comparing:

- Investment Grade versus High Yield portfolios
- Static versus Dynamic immunization methods
- Portfolio performance during different interest rate environments
- Funding stability and surplus behavior over time

To complete the analysis, two immunized fixed income portfolios were constructed using Bloomberg Treasury benchmarks and corporate bond indexes. One portfolio emphasized investment grade credit exposure, while the other included higher-yielding Baa/high yield credit exposure. At inception, both portfolios were structured to meet the key requirements of duration-based immunization, including present value matching and duration alignment with the liability.

The project examined two separate immunization approaches:

- Static Immunization (No Rebalancing)
- Dynamic Immunization (Semiannual Rebalancing)

Data and Methodology

All market data were obtained from Bloomberg. The project utilized Treasury total return indexes and Bank of America credit indexes to construct immunized portfolios

1. Treasury Indexes:

Asset Class	Bloomberg Ticker	Description
Short Treasury (1-4 Years)	I03840US Index	Bloomberg U.S. Treasury 1-4 Year Total Return Index
Intermediate Treasury (3-7 Years)	LT13TRUU Index	Bloomberg U.S. Treasury 3-7 Year Total Return Index
Core Treasury (3-10 Years)	LT31TRUU Index	Bloomberg U.S. Treasury 3-10 Year Total Return Index

2. Credit Indexes:

Credit Exposure	Bloomberg Ticker	Description
Investment Grade Credit	I02787US Index	Bloomberg U.S. Long Credit Aaa Total Return Index
High Yield / Baa Credit	I02788US Index	Bloomberg U.S. Long Credit Baa Total Return Index

The primary market inputs used throughout the project included:

- Macaulay Duration
- Market Prices
- Yields
- Coupon Rates
- Total Return Performance

The liability was defined as a future obligation of:

- Liability Amount: \$100,000,000
- Liability Date: December 31, 2025
- Initial Portfolio Construction Date: December 31, 2018

The liability horizon was therefore approximately 7 years at inception.

The liability duration was assumed to decline steadily over time as maturity approach

Part 1: Static Immunization (No Rebalancing)

The static immunization results show that both the Investment Grade and High Yield portfolios initially satisfied the required immunization conditions through present value and duration matching. At inception, both portfolios were properly aligned with the liability, suggesting that they should theoretically remain capable of funding the future obligation under stable market conditions.

However, as time progressed and market conditions changed, both portfolios experienced significant duration drift. Since the portfolios were not rebalanced, their durations no longer moved in line with the declining liability duration. This created increasing mismatches between the portfolios and the liability, weakening the effectiveness of the immunization strategy over time.

The Investment Grade portfolio demonstrated more stable behavior throughout the investment horizon, with smaller fluctuations in portfolio value and surplus relative to the liability. Despite this relative stability, the portfolio still experienced a persistent deterioration in funding status after the sharp increase in interest rates during 2022. The portfolio value eventually fell below the liability present value, resulting in sustained negative surplus levels.

The High Yield portfolio generated stronger early performance due to its higher coupon income and yield advantage. During the earlier years of the horizon, the portfolio produced larger surpluses relative to the Investment Grade strategy. However, the portfolio also exhibited substantially greater volatility and larger fluctuations in surplus behavior as market conditions evolved.

Both static portfolios ultimately became underfunded as the investment horizon shortened. The widening duration gaps and increasing surplus volatility demonstrate that duration matching at inception alone is insufficient to maintain reliable long-term immunization. Without periodic adjustments, the portfolios gradually lost their ability to effectively hedge the liability against changing interest rates and credit market conditions.

Overall, the static immunization analysis highlights the limitations of passive liability management strategies. Although the portfolios initially satisfied immunization requirements, the absence of rebalancing reduced the reliability of the hedge over time and increased exposure to funding risk.

Part 2: Dynamic Immunization (Semi-Annual Rebalancing)

The dynamic immunization results show that periodic rebalancing improved the effectiveness of both the Investment Grade and High Yield portfolios. By adjusting allocations every six months, the portfolios maintained closer alignment with the declining liability duration and reduced the duration drift observed under the static strategy.

Compared with the static portfolios, the rebalanced portfolios experienced smaller duration gaps, more stable surplus behavior, and improved liability tracking throughout the investment horizon. Although both portfolios benefited from rebalancing, the Investment Grade portfolio demonstrated a smoother and more stable adjustment process, reflecting the lower volatility and greater predictability of higher-quality credit instruments.

The High Yield portfolio required more active adjustments and larger allocation changes due to higher credit spread volatility and greater sensitivity to changing market conditions. As the liability approached maturity, the portfolio moved into cash positions earlier and more aggressively than the Investment Grade strategy.

Overall, the dynamic immunization strategy proved more effective at maintaining portfolio stability and reducing funding mismatch risk over time.

Part 3 Rebalancing

The rebalancing results show that both portfolios were adjusted over time to maintain immunization as duration declined toward zero. However, the Investment Grade portfolio experienced a smoother and more stable rebalancing process, while the High Yield portfolio required larger and more frequent allocation of changes. This reflects the effect of credit quality on portfolio stability and immunization reliability.

The Investment Grade portfolio transitioned gradually into cash positions, suggesting more predictable cash flows and lower implementation risk. In contrast, the High Yield portfolio required earlier and more aggressive reallocations into liquidity positions as the horizon shortened. These adjustments increased transaction costs, liquidity concerns, and overall portfolio management complexity.

Both portfolios also began moving into cash before the investment horizon fully matured, particularly the High Yield portfolio. This suggests that maintaining immunization became more difficult over time, forcing the portfolio manager to prioritize stability and liquidity over additional return generation.

Overall, the analysis highlights the tradeoff between higher yield and portfolio stability. The High Yield portfolio offered greater return potential but required more active management and carried greater credit, liquidity, and rebalancing risk. The Investment Grade portfolio provided a more stable and practical immunization strategy for liability-driven investing.

Conclusion

The findings of this project indicate that effective liability-driven investing cannot rely solely on initial duration matching. Even though both the Investment Grade and High Yield portfolios were properly immunized at the start of the analysis, the static strategies became progressively less effective as interest rates changed and the liability horizon shortened. Over time, this caused growing duration mismatches, increased surplus volatility, and a decline in overall funding stability.

The dynamically managed portfolios produced more consistent results throughout the investment horizon. By rebalancing the portfolios every six months, duration alignment with the liability was continuously restored, allowing the portfolios to maintain closer tracking relative to the liability value. Compared with the static strategies, the rebalanced portfolios experienced smaller funding shortfalls and more stable surplus behavior during periods of market volatility.

The project also illustrates the balance between pursuing higher returns and maintaining portfolio stability. The High Yield portfolio benefited from higher coupon income and stronger early-period performance, but it also experienced greater volatility, larger drawdowns, and more demanding rebalancing requirements. The Investment Grade portfolio, while offering lower yield potential, provided more stable liability tracking and a smoother immunization process over time.

In conclusion, the analysis suggests that dynamic immunization is a more reliable strategy for managing fixed income liabilities in changing market environments. Although higher-yield credit exposure can enhance return potential, maintaining long-term liability protection depends heavily on active duration management, periodic rebalancing, and effective risk control.

Extra Credit (Python)

To further evaluate the effectiveness of the immunization strategies, Python was used to generate visual analyses of portfolio behavior throughout the investment horizon. Historical market data, including prices, durations, yields, and index returns, were imported directly from Bloomberg into Python and used to analyze the performance of both the static and dynamic portfolios. The following figures illustrate duration drift, liability tracking performance, and surplus stability for the Investment Grade and High Yield strategies over time.

Part 1 Static Immunization (No rebalancing)

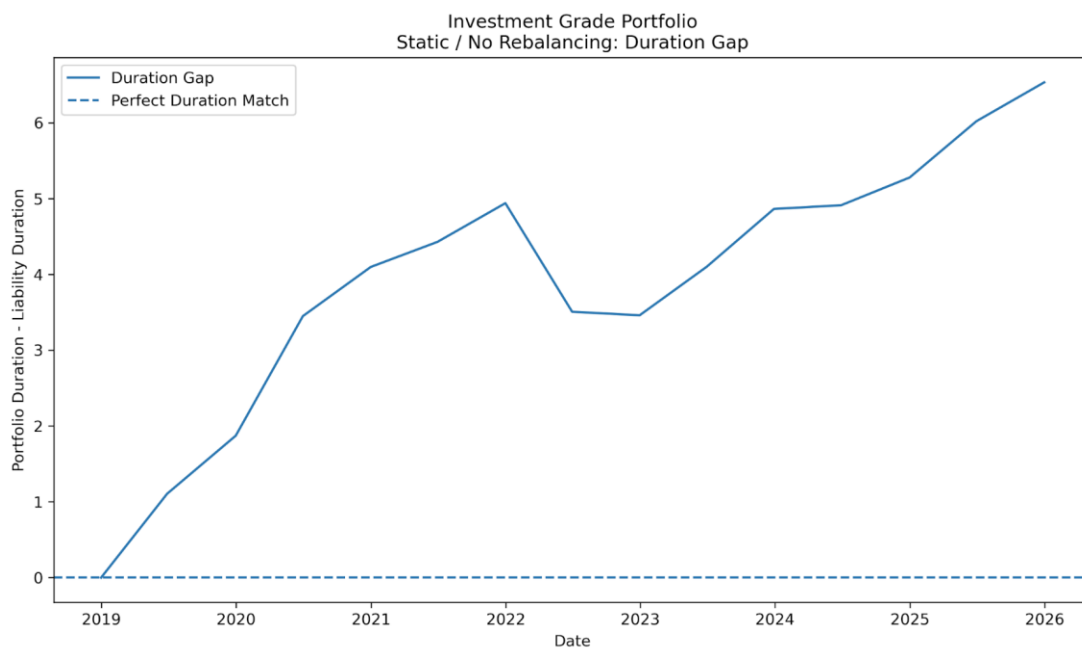


Figure 1: Illustrates the duration gap for the static Investment Grade portfolio over the investment horizon. Although the portfolio was initially immunized, the duration mismatch widened steadily over time as the liability duration declined, and the portfolio was not rebalanced. This demonstrates that static immunization becomes less effective over long horizons due to duration drift

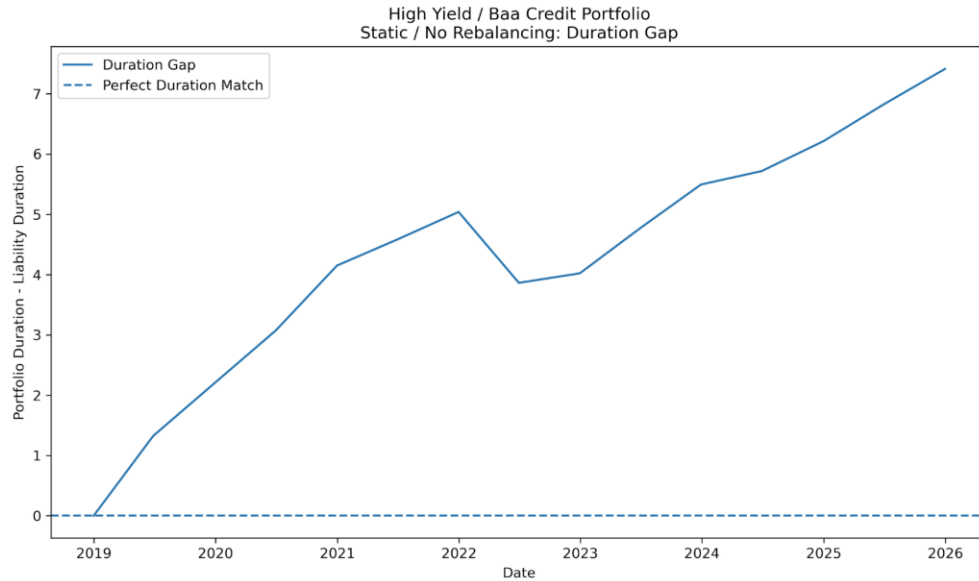


Figure 2: Shows the duration gap for the static High Yield portfolio. Similar to the Investment Grade strategy, the portfolio experienced increasing duration mismatch over time, although the fluctuations were larger due to higher credit spread volatility. The results indicate that the High Yield portfolio was more sensitive to changing market conditions and therefore less stable under a static immunization approach

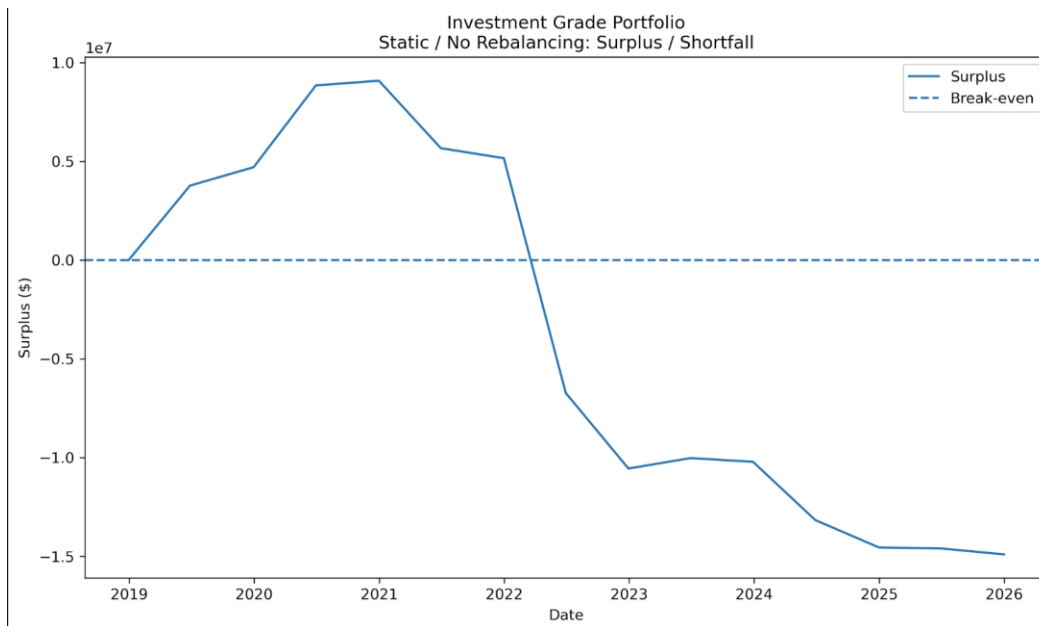


Figure 3: Illustrates the surplus behavior of the static Investment Grade portfolio throughout the investment horizon. The portfolio initially maintained a positive surplus, but funding stability deteriorated over time as duration drift increased. By the later years of the analysis, the portfolio consistently exhibited negative surplus levels, indicating underfunded status.

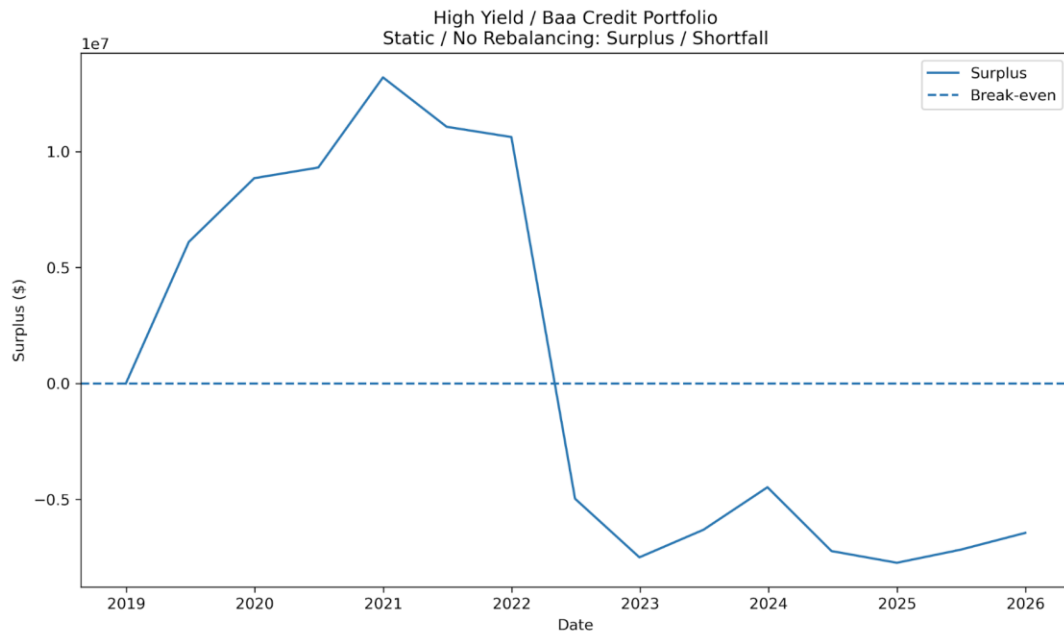


Figure 4 shows the surplus and shortfall behavior of the static High Yield portfolio. Although the portfolio generated larger early surpluses than the Investment Grade strategy, it also experienced significantly greater volatility throughout the horizon. The results demonstrate the tradeoff between higher yield generation and increased funding instability.

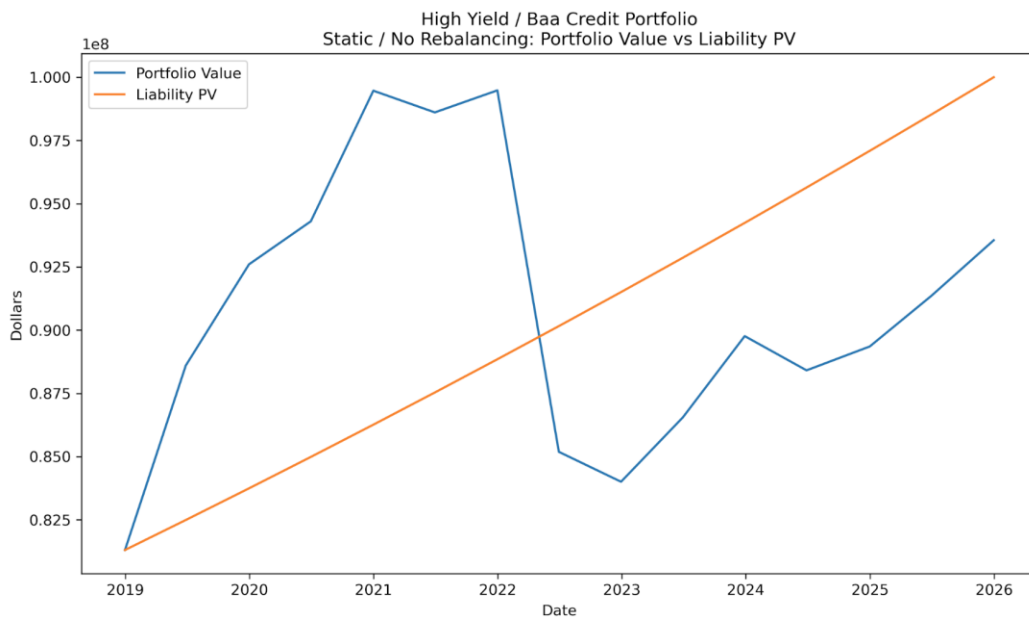


Figure 5 compares the value of the static High Yield portfolio with the present value of the liability over the investment horizon. Although the portfolio initially generated strong performance due to higher coupon income and yield advantage, its value became increasingly volatile as market conditions changed. After the 2022

interest rate shock, the portfolio fell below the liability value and struggled to fully recover, demonstrating the limitations of static immunization without periodic rebalancing.

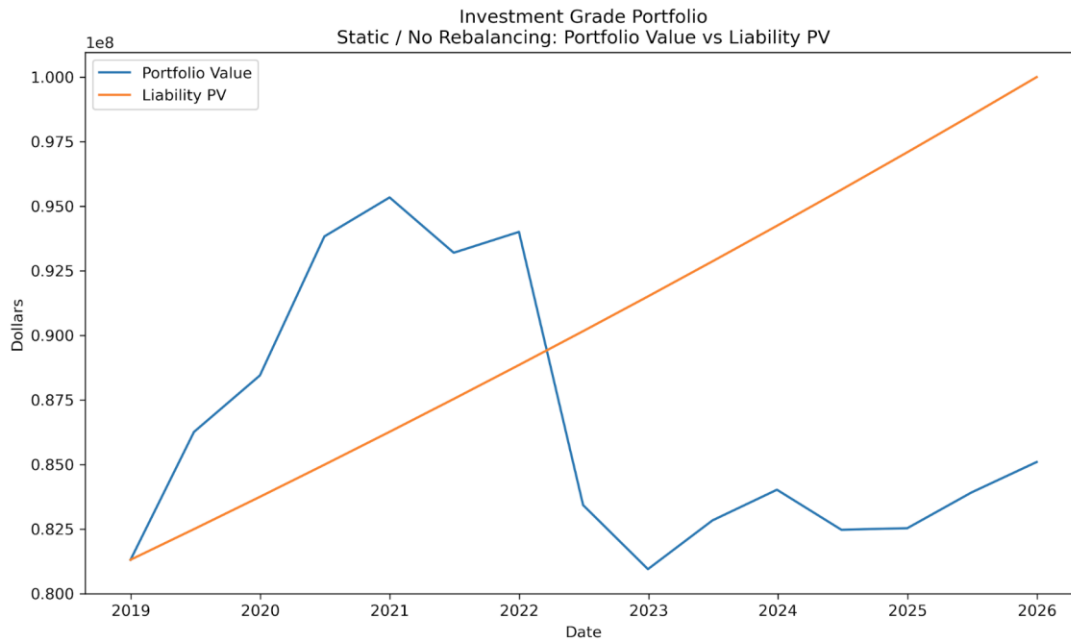


Figure 6 illustrates the relationship between the static Investment Grade portfolio value and the present value of the liability throughout the investment horizon. Although the portfolio initially tracked the liability relatively closely, its value declined significantly after the 2022 interest rate shock and remained below the liability value for the remainder of the period. This demonstrates how static immunization becomes less effective over time when portfolio allocations are not periodically rebalanced.

Part 2: Dynamic Immunization (Semi-Annual Rebalancing)

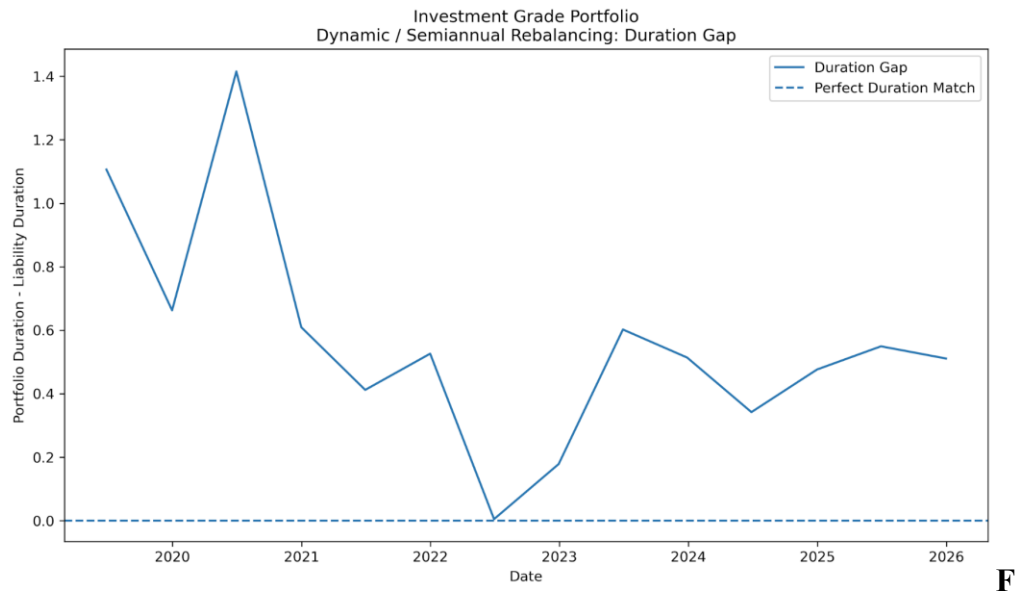


Figure 7: illustrates the duration gap for the dynamically rebalanced Investment Grade portfolio. Compared with the static strategy, the duration mismatch remained significantly smaller throughout the investment horizon due to periodic portfolio adjustments. This demonstrates the effectiveness of rebalancing in maintaining duration alignment with the liability

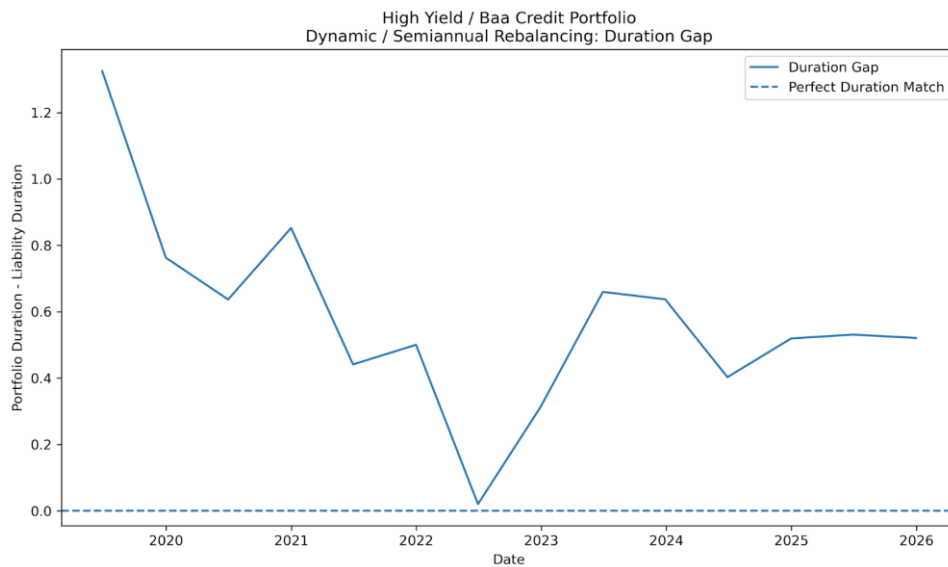


Figure 8: Presents the duration gap for the dynamically rebalanced High Yield portfolio. Although the portfolio still experienced fluctuations due to higher credit spread volatility, periodic rebalancing helped reduce the magnitude of the duration mismatch relative to the static strategy. The results show that active management improved immunization stability over time.

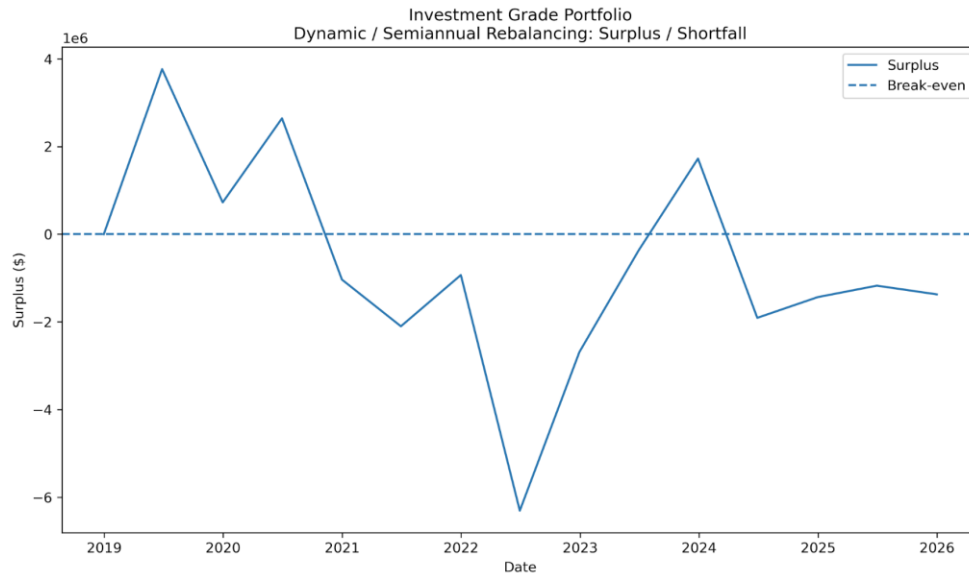


Figure 9: Shows the surplus behavior of the dynamically rebalanced Investment Grade portfolio. Compared with the static strategy, surplus fluctuations were smaller, and funding stability improved significantly throughout the investment horizon. The results indicate that active duration management reduced funding mismatch risk over time.

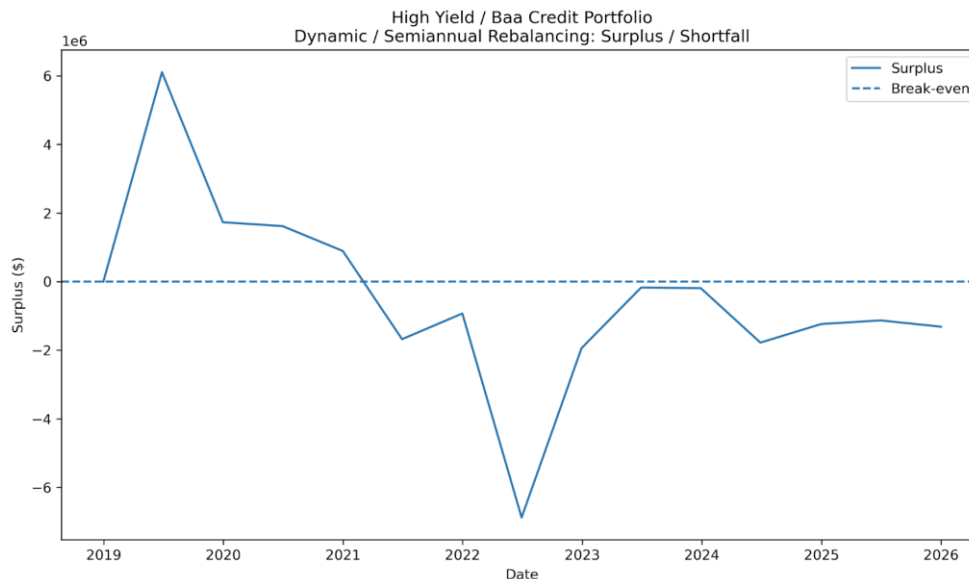


Figure 10: Presents the surplus and shortfall behavior of the dynamically rebalanced High Yield portfolio. Although temporary shortfalls still occurred during periods of market volatility, the portfolio maintained more stable funding behavior relative to the static strategy. This demonstrates that periodic rebalancing improved the overall effectiveness of the immunization process

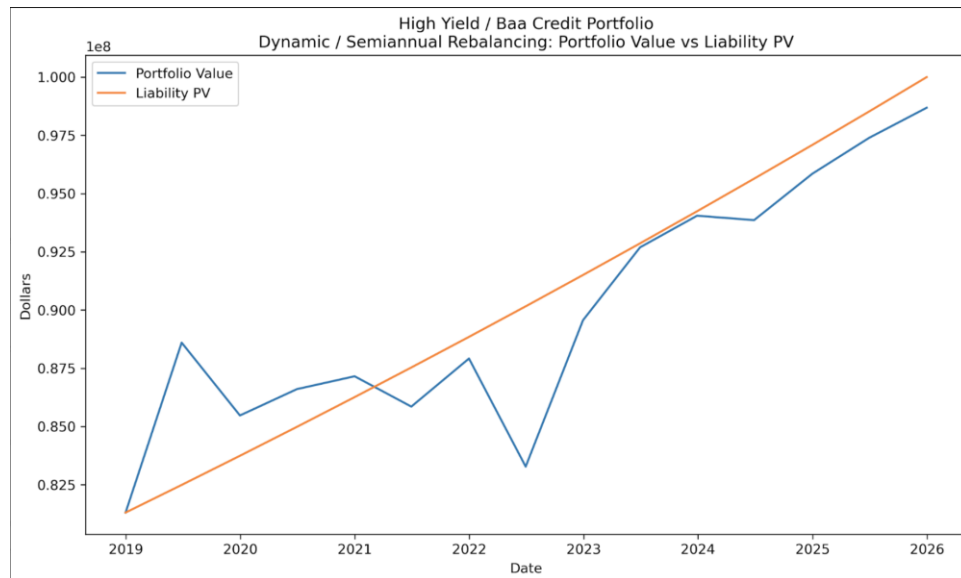


Figure 11: Compares the value of the dynamically rebalanced High Yield portfolio with the present value of the liability. Although the portfolio experienced periods of volatility, periodic rebalancing helped maintain closer tracking relative to the liability throughout the investment horizon. Compared with the static strategy, the portfolio remained closer to fully funded status and demonstrated improved funding stability over time.

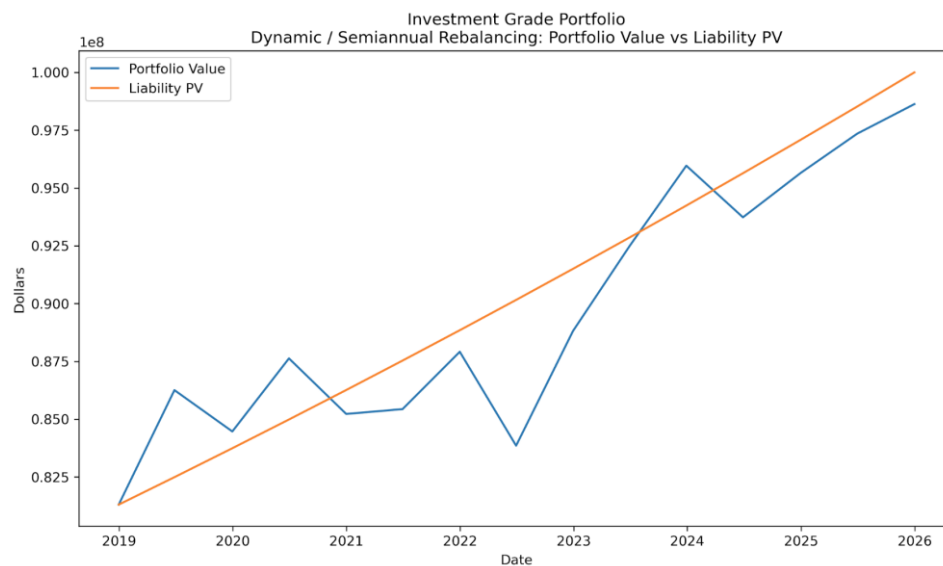


Figure 12: Compares the dynamically rebalanced Investment Grade portfolio value with the present value of the liability over the investment horizon. Although the portfolio experienced temporary fluctuations during periods of market volatility, periodic rebalancing helped maintain closer alignment with the liability compared with the static strategy. The portfolio remained closer to fully funded status throughout the horizon, demonstrating the effectiveness of active duration management and semiannual rebalancing.

Conclusion: Funding Ratio Comparison: Static vs Dynamic Immunization Strategies

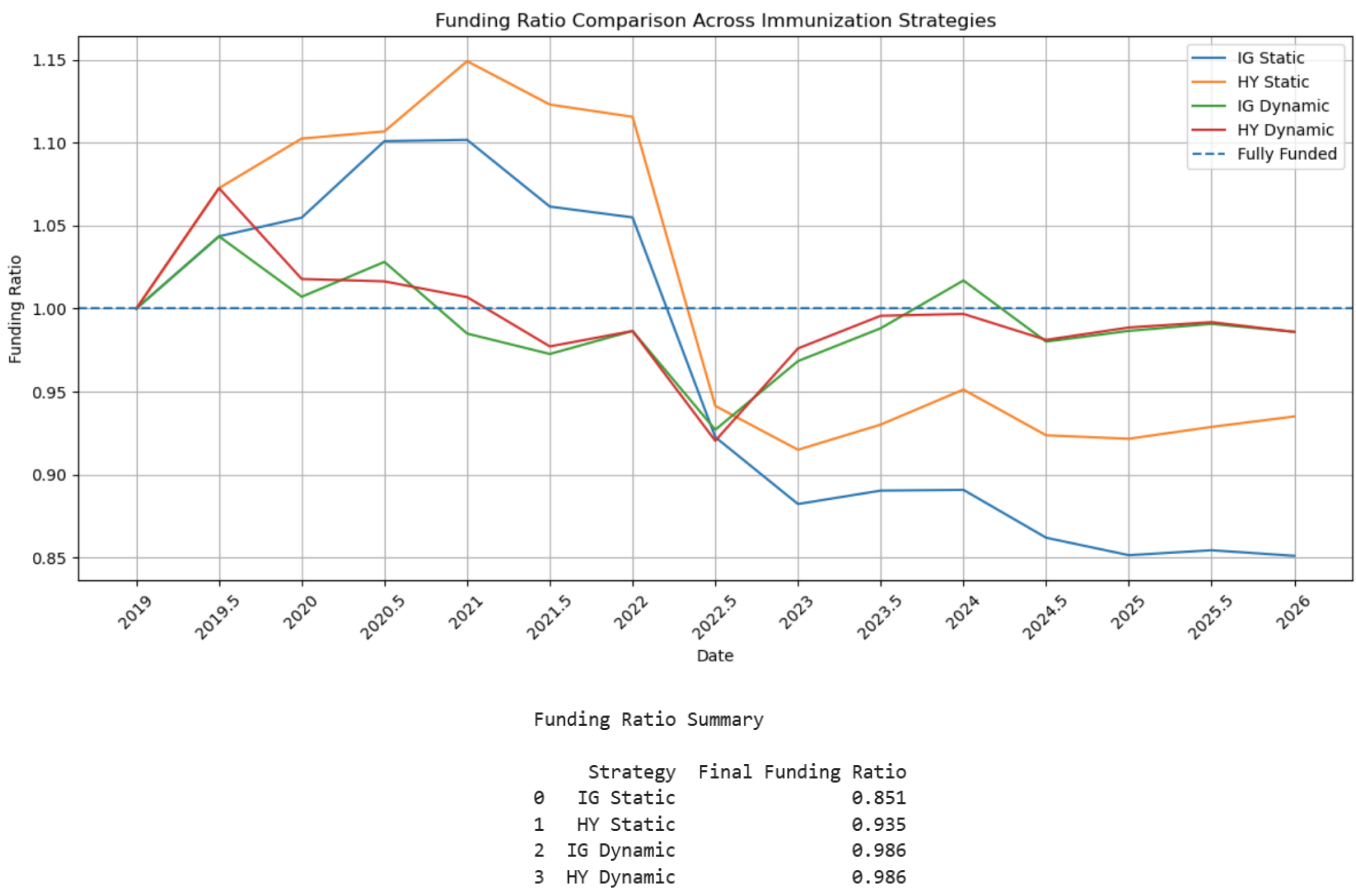


Figure 13: Represents the funding ratio comparison across all four immunization strategies. The dynamically rebalanced portfolios maintained funding ratios much closer to 1.00 throughout the investment horizon, indicating stronger liability tracking and improved funding stability. In contrast, the static portfolios experienced significant deterioration after 2022, demonstrating the limitations of passive immunization under changing market conditions.